RESULTSPRO ECOSYSTEM • BUILDEROS

CoursesPRO • Enterprise Testing & Quality Assurance Specification

Complete Verification Handbook Across All 40 Next.js Screens, Go Microservice Endpoints, WebRTC Live Video, and Multi-Tenant Subdomains

Target Application
CoursesPRO (BuilderOS LMS & Peer Learning Platform)

Frontend Architecture
Next.js 15 (React 19), TanStack Query v5, Tailwind CSS, Framer Motion

Backend Service
service_coursespro (Go 1.23, Gin, GORM, Postgres / Neon)

Identity & Authentication
service_users (JWT Bearer Token, X-Tenant-Domain resolution)

Live Video & Coworking
Daily.co WebRTC Video Call Engine with Custom Grid Layout

Document Version & Date
v2.4.0 • November 2026 Edition

Audience & Scope: This testing specification is formulated for Quality Assurance (QA) engineers, full-stack developers, security auditors, and release leads responsible for certifying the multi-tenant CoursesPRO platform prior to production deployment.

© 2026 ResultsPRO Suite • ReedBreed Creative Capital

📋 Table of Contents

SECTION	MODULE / FOCUS AREA	KEY ARCHITECTURAL COMPONENTS	TARGET ROLES
1.0	**System Architecture & Multi-Tenancy**	Subdomain routing, Tenant context, JWT auth cookies, CORS	All / Platform Engineers
2.0	**Test Personas & Seed Credentials Matrix**	Seeded students, mentors, school admins, superadmins	QA Engineers
3.0	**Public Campus & Admissions Funnel (7 Screens)**	Showcase, Cohort catalog, Student/Mentor application funnels, Store, Auth	Prospective Students & Mentors
4.0	**Student Learning Portal & Builder OS (15 Screens)**	Journey modules, Compiler tasks, Capstones, Kanban, Daily.co video	Enrolled Students / Builders
5.0	**Mentor Studio & Evaluation Console (6 Screens)**	Reviews queue, Cohort supervisions, Office hours, NGN Payout ledger	Technical Instructors / Mentors
6.0	**Academy Admin & Curriculum Builder (10 Screens)**	Visual program builder, AI quiz generator, Admissions, Payments, Gamification	School & Academy Admins
7.0	**Platform SuperAdmin & Operations (2 Screens)**	Tenant provisioning, platform subscriptions, multi-app linking	Platform SuperAdmins
8.0	**End-to-End Core User Journeys**	Step-by-step lifecycles: Enrollment → Learning → Review → Graduation	E2E Automation & Manual QA
9.0	**Complete API Endpoints Verification Matrix**	Full matrix of 30+ Go backend endpoints with request/response specs	Backend QA & Integration
10.0	**Security, Multi-Tenant Isolation & Chaos QA**	SQL injection, cross-tenant data leakage, token hijacking, concurrency	Security Auditors & DevOps

Testing Pre-requisites: Ensure local services are running on their designated ports:

- Frontend Webapp: `http://localhost:3006` (e.g. `http://skillupacademy.localhost:3006`)
- Backend Microservice (`service_coursespro`): `http://localhost:8080`
- Identity Service (`service_users`): `http://localhost:8081`
- Database: Neon PostgreSQL with seeded tenant `skillupacademy` and `greenwood-high`

1.0 System Architecture & Multi-Tenancy QA

CoursesPRO is a multi-tenant application that dynamically resolves institutional brand identities based on the request domain. Correct multi-tenant resolution is essential before running any UI or API test.

1.1 Tenant Resolution Architecture

`lib/api.ts → getTenantSlug()`

The frontend extracts the tenant identifier from the hostname:

- **Production Subdomains:** `skillupacademy.resultspro.ng` → Tenant: `skillupacademy`
- **Local Development Subdomains:** `skillupacademy.localhost:3006` → Tenant: `skillupacademy`
- **Plain Localhost / Platform Root:** `localhost:3006` → Falls back to `NEXT_PUBLIC_TENANT_SLUG`
- **Custom Domains:** `learn.skilluplearningacademy.com` → Full canonical host mapped in tenant registry

> **Header Injection:** Every HTTP call generated by Axios includes the `X-Tenant-Domain` request header. The Go backend middleware reads this header, resolves the `tenant_id` from the database, and injects `db.WithTenant(c)` into all GORM queries.

2.0 Test Personas & Seed Credentials Matrix

Use the following pre-seeded test accounts to validate role-based permissions and access control:

PERSONA ROLE	EMAIL ADDRESS	PASSWORD	TENANT DOMAIN	PRIMARY PERMISSIONS
Platform SuperAdmin	`superadmin@resultspro.ng`	`Password123!`	Root / Platform	All tenant configurations, billing, global analytics
Academy Admin	`tenant-admin@example.edu`	`Password123!`	`skillupacademy`	Curriculum builder, cohorts, student approvals, earnings
Lead Mentor / Instructor	`usr_mentor_001@example.com`	`Password123!`	`skillupacademy`	Project code reviews, grading, live classroom, payout requests
Associate Mentor	`usr_mentor_002@example.com`	`Password123!`	`skillupacademy`	Assigned cohort reviews, 1:1 office hours booking link
Enrolled Student (Active)	`student@example.com`	`Password123!`	`skillupacademy`	Stage 1–3 lessons, compiler challenges, capstone submissions
Prospective Applicant	`newstudent@test.com`	`Password123!`	`skillupacademy`	Public catalog, admissions application, tuition payment

3.0 Public Campus & Admissions Funnel (Screens 1 – 7)

Screen 01 • Tenant Campus Home

`/[tenant] • Public`

Purpose: Brand landing page displaying active tracks, academy statistics, student testimonials, and tuition fees.

- [ ] **Branding:** Verify logo, academy colors, and motto match tenant settings in `crs_tenant_settings`.
- [ ] **Hero Action:** Click "Explore Tracks" → smooth scrolls or navigates to `/[tenant]/cohorts`.
- [ ] **Testimonials:** Validate builder project showcases and testimonials render without layout distortion.

Screen 02 • Cohort Selection & Catalog

`/[tenant]/cohorts • Public`

Purpose: Displays active batches with start dates, meeting days, difficulty ratings, and enrollment status.

- [ ] **Data Grid:** Verify only cohorts with `status = 'ACTIVE'` are displayed.
- [ ] **Enroll CTA:** Clicking "Apply Now" passes the selected `cohort_id` query param to `/[tenant]/apply?cohort={id}`.

Screen 03 • Student Application Funnel

`/[tenant]/apply • Public / Guest`

Purpose: Admissions application form collecting background, track preference, technical experience, and goals.

- [ ] **Validation:** Submit empty form → ensure required field errors appear on name, email, and goals.
- [ ] **Submission:** Fill valid applicant info → verify record creation in `crs_enrollments` with status `APPLIED`.
- [ ] **Redirection:** Ensure user is routed to `/[tenant]/onboarding/plan` upon successful submission.

Screen 04 • Mentor Application Funnel

`/[tenant]/apply/mentor • Public`

Purpose: Instructor application form collecting GitHub, LinkedIn, technical specialization, and sample code review links.

- [ ] **Submission:** Submit mentor application → check record inserted into `crs_mentor_applications` with status `PENDING_REVIEW`.
- [ ] **Admin Notification:** Verify application shows in `/admin/mentors` under Pending Applicants.

Screen 05 • Public Campus Store

`/[tenant]/store • Public`

Purpose: Public catalog of academy merchandise, hoodies, textbooks, and workshop tickets.

- [ ] **Currency Display:** Confirm prices display properly in Nigerian Naira (`₦`) with thousand separators.
- [ ] **Inventory Check:** Out-of-stock items should display a disabled "Sold Out" state.

Screen 06 • Public Student Portfolio Showcase

`/[tenant]/portfolio/[username] • Public`

Purpose: Verified proof-of-work portfolio displaying student capstones, verified certificates, and GitHub links.

- [ ] **Live Rendering:** Navigate to `/skillupacademy/portfolio/janedoe` → verify profile bio, achievements, and approved projects render.
- [ ] **Security:** Verify unapproved drafts or private notes are NOT exposed in the public view.

Screen 07 • Authentication & Onboarding

`/[tenant]/login, /signup, /onboarding/* • Auth`

Purpose: User sign-in, account registration, email verification, plan selection, and campus orientation.

- [ ] **JWT Auth Cookie:** Upon login, check browser cookies for `token` scoped to the platform domain.
- [ ] **Role Redirect:** Mentors redirect to `/mentor`, Admins to `/admin`, Students to `/dashboard`.

4.0 Student Learning Portal & Builder OS (Screens 8 – 22)

Screen 08 • Student Dashboard Overview

`/[tenant]/dashboard • Student Role`

Purpose: Home view showing current milestone stage, daily streak, XP balance, next live class, and recent submissions.

- [ ] **KPI Cards:** Validate XP count, Current Stage number, Streak counter, and Completed Modules count.
- [ ] **Quick Action:** Click "Continue Journey" → navigates to active module in `/dashboard/journey`.

Screen 09 • Interactive Journey Stages

`/[tenant]/dashboard/journey • Student Role`

Purpose: Visual roadmap of program stages (e.g. Stage 1: Fundamentals, Stage 2: React, Stage 3: Fullstack Backend).

- [ ] **Milestone Locking:** Stages higher than `current_stage_number` should display locked with requirements.
- [ ] **Progress Rings:** Percentage progress recalculates dynamically as module blocks are checked off.

Screen 10 • Interactive Lesson & Module Workspace

`/[tenant]/dashboard/journey/[moduleId] • Student Role`

Purpose: Core learning workspace containing lesson markdown, video lectures, coding compiler, quizzes, and reflection blocks.

- [ ] **Video Player:** Embeds YouTube/Vimeo/Cloudinary video player with time tracking.
- [ ] **Code Compiler:** Execute Javascript/Go/Python snippet → verify stdout renders in terminal pane.
- [ ] **Quiz Validation:** Complete multiple-choice quiz → score calculates instantly; passing marks module progress.
- [ ] **Block Submission:** Submit assignment block (code or text) → verify `POST /api/submissions/blocks` returns `201 Created`.

Screen 11 • Capstone Projects Studio

`/[tenant]/dashboard/projects • Student Role`

Purpose: Submitting end-of-stage capstone projects with GitHub repo, live demo URL, and student architectural notes.

- [ ] **Submission Form:** Enter GitHub repo URL, Figma URL, demo URL → submit via `POST /api/projects/submit`.
- [ ] **Status Indicator:** Project status shifts to `MENTOR_REVIEW` badge.
- [ ] **Feedback Drawer:** After review, instructor rating (1–5 stars), written critique, and Loom video URL render.

Screen 12 • Live Coworking & Classroom Studio

`/[tenant]/dashboard/classroom • Student & Mentor`

Purpose: Integrated Daily.co WebRTC live video room for sprint kickoff, workshops, and office hours.

- [ ] **WebRTC Connection:** Room joins without token failure; camera and microphone toggles respond smoothly.
- [ ] **Screen Share:** Share screen → video feeds adjust to presentation grid.
- [ ] **Presence Heartbeat:** Verify `POST /api/classroom/heartbeat` fires every 30s to update presence in `crs_presence_sessions`.

Screen 13 • Student Kanban Workspace

`/[tenant]/dashboard/workspace • Student Role`

Purpose: Drag-and-drop task management board for planning capstone sprints and feature deliverables.

- [ ] **Task Creation:** Click "+ New Task", fill title, due date & priority → task card appears in "Backlog" column.
- [ ] **Drag & Drop:** Move task from "In Progress" to "Completed" → `PUT /api/workspace/tasks/:id/status` updates state.

Screens 14 – 22 • Social, Gamification & Campus Utilities

`/[tenant]/dashboard/(peers, messages, leaderboard, achievements, certificates, store, billing, portfolio, mentor, resources)`

SCREEN / ROUTE	PRIMARY ACTION	VALIDATION CRITERIA
`/dashboard/peers`	Pair programming roster	Displays active cohort peers with status badges (Online/Offline)
`/dashboard/messages`	1:1 Direct Chat	Send direct message → delivered in real time via `POST /api/messages/conversations/:id`
`/dashboard/leaderboard`	Weekly XP ranking	Sorts students by `current_xp` with gold, silver, and bronze trophies
`/dashboard/achievements`	Badges & unlocks	Displays earned badges (e.g. `FIRST_PROJECT_APPROVED`) with unlock timestamps
`/dashboard/certificates`	Verified graduation certs	Generates verifiable PDF/PNG credential with unique credential ID and verification link
`/dashboard/store`	XP redemption & merch	Allows redeeming campus swag using accumulated XP; deducts coins from balance
`/dashboard/billing`	Tuition & Paystack gateway	Initiates payment intent, opens Paystack popup, verifies webhook enrollment
`/dashboard/portfolio`	Portfolio CMS	Custom bio, featured project toggles, custom social handles update in `crs_public_portfolios`
`/dashboard/mentor`	Book 1:1 office hours	Direct link opens instructor's Calendly/Cal.com booking modal
`/dashboard/resources`	Curated technical library	Searchable cheat sheets, starter templates, and design system kits

5.0 Mentor Studio & Evaluation Console (Screens 23 – 28)

Screen 23 • Mentor Overview Dashboard

`/mentor • Mentor Role`

Purpose: Mission control showing pending review count, upcoming live sessions, assigned cohorts, and quick grade action.

- [ ] **Queue Card:** Verifies dynamic pending review count matches backend `GET /api/mentor/submissions`.
- [ ] **"Grade Now" CTA:** Routes directly to `/mentor/reviews`.

Screen 24 • Code Review Evaluation Studio

`/mentor/reviews • Mentor Role • ZERO MOCKS CERTIFIED`

Purpose: Comprehensive grading studio with dual tabs for Capstone Projects and Module Block Tasks.

- [ ] **Tab Switching:** Toggle between "Capstone Projects" and "Module Assignments & Tasks"; badges reflect accurate counts.
- [ ] **GitHub & Demo Inspection:** Click GitHub / Demo buttons → opens student repos and preview apps in new tabs.
- [ ] **Rating & Video URL:** Select 1–5 stars; provide written critique; optionally add Loom review link.
- [ ] **Revision Flow:** Click "Request Revision" → submission status changes to `REVISION_REQUESTED`.
- [ ] **Approval Flow:** Click "Approve & Award XP" → student advances stage, awards +500 XP, updates mentor rating.
- [ ] **Block Grading:** Select Module Task → enter score (0–100), feedback → submits to `POST /api/mentor/submissions/blocks/:id/review`.

Screen 25 • Assigned Cohorts & Mentee Supervision

`/mentor/cohorts • Mentor Role • ZERO MOCKS CERTIFIED`

Purpose: Manage assigned student batches, inspect progress milestones, and launch live coworking studios.

- [ ] **Search Filtering:** Real-time search filters cohorts and student rosters by title, program, or student name/email.
- [ ] **Expandable Roster:** Click "View Mentees" → displays enrolled students with avatars, stage, XP, and streak.
- [ ] **Direct Message Button:** Click "Direct Message" → routes directly to `/dashboard/messages?user={student.id}`.

Screen 26 • Live Sessions, Workshops & Office Hours

`/mentor/sessions • Mentor Role • ZERO MOCKS CERTIFIED`

Purpose: Host live classes, review recurring schedules, and share 1:1 Cal.com / Calendly booking links.

- [ ] **Clipboard Copy:** Click "Copy Booking Link" → copies Cal.com URL; animates "Copied to Clipboard!" checkmark.
- [ ] **Temporal Tabs:** Validate "Scheduled Workshops" lists upcoming sessions; "Past Sessions" filters historical dates.
- [ ] **Studio Join CTA:** Click "Open Live Classroom Studio" → routes to Daily.co room in `/dashboard/classroom`.

Screen 27 • Mentor Compensation & Escrow Ledger

`/mentor/earnings • Mentor Role • ZERO MOCKS CERTIFIED`

Purpose: Financial ledger tracking evaluation bounties (₦15,000/review), base stipends, pending escrow, and NGN bank withdrawals.

- [ ] **Escrow & Balance:** Verify Cleared Balance reflects Gross Earnings minus Withdrawals; Pending Escrow tracks unreviewed projects.
- [ ] **Withdrawal Modal:** Click "Withdraw Funds" → select Nigerian bank (GTBank, Access, Zenith, OPay, Moniepoint, etc.).
- [ ] **NUBAN Validation:** Enter 10-digit account number → rejects inputs under 10 digits or amount below ₦5,000.
- [ ] **Payout Mutation:** Submit → calls `POST /api/mentor/payouts/request`; updates ledger with status `PENDING`.

Screen 28 • Mentor Settings & Showcase

`/mentor/settings • Mentor Role • ZERO MOCKS CERTIFIED`

Purpose: Configure professional identity, avatar, domain specialization, and personal booking URL.

- [ ] **Profile Update:** Modify specialization or avatar URL → submit → calls `PUT /api/mentor/profile` → shows success toast.
- [ ] **Booking URL Test:** Enter Cal.com link → "Test Your Booking Page" link opens external booking site correctly.

6.0 Academy Admin & Curriculum Builder (Screens 29 – 38)

Screen 29 • Academy Admin Dashboard

`/admin • SchoolAdmin Role`

Purpose: High-level institutional metrics: Active Cohorts, Enrolled Builders, Course Completion Rate, Gross Tuition Collected.

- [ ] **Metrics Accuracy:** Verify counters match active database records across cohorts, enrollments, and transactions.
- [ ] **Quick Navigation:** Verify links to Program Builder, Cohorts, and Student Roster operate correctly.

Screens 30 & 31 • Visual Program & Curriculum Builder

`/admin/program-builder & /admin/program-builder/[id] • Admin`

Purpose: Authoring studio for creating multi-stage curriculums, modules, lessons, coding challenges, and AI-generated quizzes.

- [ ] **Program CRUD:** Create new program with title, description, base tuition fee & duration in weeks.
- [ ] **Stage Drag & Drop:** Add stages (Order index 1, 2, 3) → rearrange order → verify order persisted in `crs_journey_stages`.
- [ ] **Module Editor:** Add video lectures, rich markdown content, and attach compiler coding challenges.
- [ ] **AI Quiz Generator:** Input lesson text → trigger Gemini AI quiz generation → generates 5 Bloom-taxonomy multiple-choice questions automatically into `crs_quizzes`.

Screens 32 – 38 • Academy Operations & Management

`/admin/(cohorts, students, mentors, payments, store, achievements, reports, settings)`

ADMIN SCREEN	KEY FEATURES & ACTIONS	EXPECTED QA RESULT
`/admin/cohorts`	Cohort creation, date scheduling, meeting days, capacity limits	New cohort appears in student enrollment catalog
`/admin/students`	Student enrollment roster, manual stage advancement, expulsion/status toggle	Updates `crs_enrollments` status (`ACTIVE` , `PAUSED` , `GRADUATED`)
`/admin/mentors`	Mentor application review, cohort mentor assignments (Lead/Associate)	Assigns mentor to cohort in `crs_cohort_mentors`
`/admin/payments`	Tuition reconciliation, transaction auditing, refund management	Displays Paystack references, amounts, and settlement statuses
`/admin/store`	Manage campus merchandise, stock inventory, XP pricing	Item updates reflect in student store catalog
`/admin/achievements`	Badge builder, code name triggers (e.g. `SPEED_DEMON` , `PERFECT_STREAK`)	Badges register in `crs_achievements` rule evaluation engine
`/admin/reports`	Completion rates, stage dropoff analytics, mentor SLA compliance	Generates cohort performance charts and exportable CSVs
`/admin/settings`	Tenant logo, branding colors, payout model (`BASE_PLUS_SLA`)	Updates reflect in `crs_tenant_settings` and student navbar

7.0 Platform SuperAdmin & Operations (Screens 39 – 40)

Screen 39 • Platform Tenants Directory

`/platform-admin/tenants • SuperAdmin Role`

Purpose: Central directory of all institutions, coding academies, and university faculties hosted on the CoursesPRO cluster.

- [ ] **Tenant Provisioning:** Create new tenant with subdomain slug → provisions tenant schema and default settings.
- [ ] **Status Toggle:** Suspend or activate tenant → invalidates unauthorized traffic instantly with 403 Forbidden.

Screen 40 • Tenant Subscription Billing & Gateways

`/platform-admin/tenants/[id]/payments • SuperAdmin Role`

Purpose: Institutional SaaS subscription management, Paystack / Stripe recurring billings, and revenue share tracking.

- [ ] **Subscription Tier:** Upgrade tenant tier (PRO, ENTERPRISE) → unlocks expanded cohort seats and custom domain SSL.

8.0 End-to-End Core User Journeys

Journey 1: Student End-to-End Lifecycle

E2E FLOW

Step 1: Guest visits `/skillupacademy/cohorts`, chooses "Fullstack Mastery Fall 2026", and completes admissions form at `/apply`.

Step 2: Logs in with seeded credentials, views `/dashboard`, and enters active module in `/dashboard/journey/stage_001`.

Step 3: Reads lesson markdown, runs code in compiler, answers interactive quiz questions, and earns +50 XP.

Step 4: Builds Stage 1 Capstone (Personal Portfolio), pushes to GitHub, and submits at `/dashboard/projects` with demo link.

Step 5: Waits for review • receives notification • views 5-star rating and Loom review feedback in project history.

Step 6: Milestone unlocks Stage 2 automatically; XP updates on the weekly leaderboard at `/dashboard/leaderboard`.

Journey 2: Mentor Evaluation & Payout Lifecycle

E2E FLOW

Step 1: Mentor logs in → redirected to `/mentor`; pending badge displays `3 Pending`.

Step 2: Navigates to `/mentor/reviews` • inspects "Capstone Projects" backlog • clicks on student submission.

Step 3: Reviews live GitHub repository & demo link • enters 5 stars • provides written code critique • clicks "Approve & Award XP".

Step 4: Switches to "Module Assignments & Tasks" tab • grades Javascript compiler challenge • assigns 95/100 points.

Step 5: Opens `/mentor/earnings` • verifies ₦15,000 bounty credited to Cleared Balance.

Step 6: Clicks "Withdraw Funds" • selects GTBank • enters 10-digit NUBAN • confirms ₦50,000 withdrawal request.

9.0 Complete API Endpoints Verification Matrix

All endpoints are implemented in `service_coursespro` (Go 1.23, Gin framework) and authenticated with Bearer tokens:

HTTP METHOD	ENDPOINT ROUTE	GO HANDLER	DATABASE TABLE	SUCCESS CODE
GET	`/api/cohorts/:id/journey`	`h.GetCohortJourney`	`crs_journey_stages`	200 OK
POST	`/api/modules/:id/progress`	`h.UpdateModuleProgress`	`crs_module_progress`	200 OK
POST	`/api/projects/submit`	`h.SubmitProject`	`crs_project_submissions`	201 Created
GET	`/api/projects/my-submissions`	`h.GetMySubmissions`	`crs_project_submissions`	200 OK
POST	`/api/submissions/blocks`	`h.SubmitBlock`	`crs_block_submissions`	201 Created
GET	`/api/submissions/blocks`	`h.GetMyBlockSubmissions`	`crs_block_submissions`	200 OK
GET	`/api/mentor/profile`	`h.GetMentorProfile`	`crs_mentor_profiles`	200 OK
PUT	`/api/mentor/profile`	`h.UpdateMentorProfile`	`crs_mentor_profiles`	200 OK
GET	`/api/mentor/cohorts`	`h.GetMentorCohorts`	`crs_cohorts`, `crs_enrollments`	200 OK
GET	`/api/mentor/sessions`	`h.GetMentorSessions`	`crs_cohort_events`	200 OK
GET	`/api/mentor/submissions`	`h.GetPendingSubmissions`	`crs_project_submissions`	200 OK
POST	`/api/mentor/submissions/:id/review`	`h.ReviewSubmission`	`crs_project_submissions`	200 OK
GET	`/api/mentor/submissions/blocks`	`h.GetPendingBlockSubmissions`	`crs_block_submissions`	200 OK
POST	`/api/mentor/submissions/blocks/:id/review`	`h.ReviewBlockSubmission`	`crs_block_submissions`	200 OK
GET	`/api/mentor/earnings`	`h.GetMentorEarnings`	`crs_mentor_payout_requests`	200 OK
POST	`/api/mentor/payouts/request`	`h.RequestMentorPayout`	`crs_mentor_payout_requests`	201 Created
GET	`/api/classroom/presence`	`h.GetPresence`	`crs_presence_sessions`	200 OK
POST	`/api/classroom/heartbeat`	`h.PresenceHeartbeat`	`crs_presence_sessions`	200 OK
GET	`/api/peers/roster`	`h.GetPeers`	`crs_enrollments`	200 OK
GET	`/api/leaderboard`	`h.GetLeaderboard`	`crs_enrollments`	200 OK
GET	`/api/workspace/tasks`	`h.GetWorkspaceTasks`	`crs_workspace_tasks`	200 OK
POST	`/api/workspace/tasks`	`h.CreateWorkspaceTask`	`crs_workspace_tasks`	201 Created
PUT	`/api/workspace/tasks/:id/status`	`h.UpdateWorkspaceTaskStatus`	`crs_workspace_tasks`	200 OK
DELETE	`/api/workspace/tasks/:id`	`h.DeleteWorkspaceTask`	`crs_workspace_tasks`	200 OK
GET	`/api/messages/conversations`	`h.GetConversations`	`crs_conversations`	200 OK
POST	`/api/messages/conversations/:id`	`h.SendMessage`	`crs_messages`	201 Created
GET	`/api/achievements`	`h.GetMyAchievements`	`crs_user_achievements`	200 OK

10.0 Security, Multi-Tenant Isolation & Chaos QA

10.1 Multi-Tenant Data Isolation Audit

P0 CRITICAL

Threat Model: A malicious student or mentor on tenant `skillupacademy` attempting to access cohorts, project repos, or financial earnings belonging to tenant `greenwood-high`.

- ☐ **Cross-Tenant ID Spoofing:** Send a `GET /api/mentor/submissions` with `X-Tenant-Domain: skillupacademy` while passing a submission ID belonging to `greenwood-high`.
  → **Pass Criteria:** GORM `db.WithTenant(c)` query returns `404 Not Found` or empty array. No data leakage occurs.
- ☐ **Missing Tenant Header:** Send API call with no `X-Tenant-Domain` header.
  → **Pass Criteria:** Middleware executes fallback `WHERE 1 = 0`; zero database records returned.

10.2 Role-Based Access Control (RBAC) Integrity

P1 SECURITY

Threat Model: Enrolled students attempting to evaluate their own capstones or withdraw instructor funds.

- ☐ **Student Access to Review Mutations:** Log in as `student@example.com`; attempt to send `POST /api/mentor/submissions/{id}/review` with status `APPROVED`.
  → **Pass Criteria:** Server rejects request or rejects grading unassigned milestones.
- ☐ **Student Access to Payouts:** Student sends `POST /api/mentor/payouts/request`.
  → **Pass Criteria:** Fails with 400 Bad Request or rejects because cleared balance is zero.

10.3 Concurrency & Chaos Testing

P2 STABILITY

- ☐ **Concurrent Payout Requests:** Send 5 simultaneous withdrawal requests of ₦50,000 against a ₦60,000 balance.
  → **Pass Criteria:** Only 1 request succeeds; subsequent requests fail with insufficient funds.
- ☐ **Websocket / Presence Load:** Simulate 100 students sending heartbeat pings simultaneously to `/api/classroom/heartbeat`.
  → **Pass Criteria:** Go Gin server maintains <50ms response latency without database connection pool exhaustion.

Sign-off Certification: Upon successful execution of all checklists in Sections 3.0 through 10.0 with zero P0/P1 defects, sign off below to certify the CoursesPRO release build for deployment.

QA Lead Signature: _____ **Date:** _____